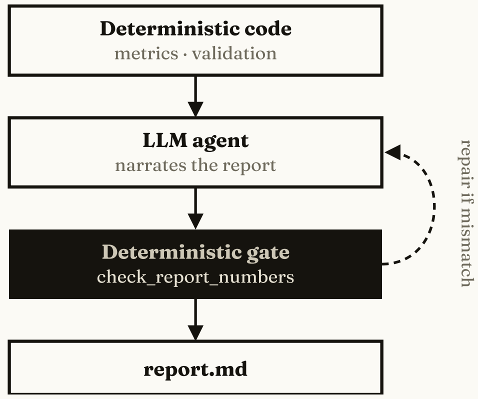
NUMBERS BY CODE, STORY BY AGENT

A reporting agent where deterministic code owns every figure and the model never does arithmetic — making it very hard for a wrong figure to reach the page.

THE PRINCIPLE

In fintech the cost of a wrong figure is high, so financial maths is not delegated to a model. Deterministic Python generates the data, computes every metric, and validates it — that is the single source of truth. The LLM agent is an **orchestrator, narrator and self-critic**: it reads the pre-computed numbers, decides which trends matter, writes the report, and verifies its own output. It is never handed raw data to "do maths" on. Either way the figures are identical, because they come from the same code — the agent only supplies prose.



Code computes and validates; the agent narrates; a deterministic gate re-checks every number before the report ships.

THE PIPELINE

1 **Generate** — a closed 1,000-user cohort over 12 months (synthetic, reproducible).

2 **Compute** — the six monthly metrics: active & paid users, churned users, revenue, churn rate, ARPU.

3 **Validate** — 17 data-quality checks gate the data before any conclusion.

4 **Report** — the agent writes a six-section business report from the validated numbers.

DATA QUALITY — 17 CHECKS

The data is distrusted until proven clean: schema, no nulls, unique (user, month), exactly 1,000 users, plan↔price, no reactivation, one lapse per user, revenue reconciliation, and more.

- **Independent recomputation** — every metric is recomputed from raw data and compared to the metrics table, so churn is verified two ways, not just totalled.
- A failed check forces a warning header and demotes conclusions to provisional.

REPRODUCIBILITY

Everything is driven by a single **seed** (**42**) via NumPy. In a pinned environment the same seed yields a byte-identical dataset and metrics — anyone who re-runs it sees the same figures. Random, but repeatable.

THE AGENT LOOP

A tool-using loop on the Anthropic Messages API. The agent receives the metrics, validation summary and pre-computed highlights, plus hard rules: **use only these numbers, never recalculate.** It drafts, calls a check tool, and revises until clean.

THE GUARDRAIL

check_report_numbers is deterministic code, not the model. It extracts every numeric token from the draft and matches it against the allowed figures — and it also flags wrong-sign percentages.

Crucially it runs on the **final** text **unconditionally** — even if the agent skipped its own check — so trust in the numbers does not depend on the model behaving.

> *In a real run the gate caught fabricated numbers on the first pass (***ok=false***), forced a repair round, and only then shipped — recorded in* **agent_transcript.md** *as evidence.*

CODE VS LLM

DETERMINISTIC CODE	LLM AGENT
All metric maths	Which trends matter
Validation & reconciliation	Business narrative
Number verification	Self-critique loop
Reproducible figures	Clear prose only